QALA

Solution Factory Operating System

Problem Requirements Document (PRD) v1.0

Field	Value
Document Type	Problem Requirements Document (PRD)
System	Qala Solution Factory Operating System
Version	1.0
Status	Draft — For Review
Purpose	Define the problems Qala solves, for whom, and how success will be measured
Companion Docs	Qala SDD v2.0 | Requirements Document v1.0 | Workflows & Use Cases v1.0
Audience	Product, Engineering, Leadership, Investors, Customers
Classification	Confidential

1 Executive Summary

Qala is a Solution Factory Operating System designed to eliminate the fragmented, error-prone, and ungoverned toolchains that plague modern software and solution delivery. It provides a unified platform that manages every stage of the solution lifecycle — from a developer's first idea to a production system serving customers — with built-in security, reproducibility, and intelligence.

The software and technology industry spends an enormous portion of its effort not on building solutions but on managing the infrastructure required to build, deliver, and govern those solutions. Teams assemble fragmented toolchains from dozens of disparate tools. Environments drift. Builds are not reproducible. Security is bolted on rather than built in. Knowledge is siloed. Governance is manual. AI intelligence is available in theory but never coherently integrated into the delivery lifecycle.

Qala solves these problems by providing a single, opinionated, extensible operating system for solution factories — whether that factory is a solo developer building their first app or a global enterprise managing hundreds of microservices across dozens of teams.

> *Qala's core thesis: every person or organisation that builds something of value deserves the same world-class delivery infrastructure that only the largest technology companies currently have. Qala makes this available to everyone.*

Dimension	The Problem Today	Qala's Answer
Build Reproducibility	Builds break on different machines. "Works on my machine" is endemic.	Hermetic builds: every build is fully isolated and produces identical outputs from identical inputs.
Security	Security is manual, late, and reactive. Vulnerabilities discovered post-deployment.	Security embedded at every layer: SAST, SCA, DAST, SEM, threat modelling, and signed attestations — all integrated.
Environment Drift	Production and development environments diverge silently. Incidents caused by undocumented configuration changes.	Environment-as-code with continuous drift detection and governed change control for every configuration change.
Knowledge Silos	Best practices live in individuals' heads. Onboarding takes months. Prior art is lost.	All solutions, artefacts, toolchains, and decisions are version-controlled, searchable, and AI-augmented.
Governance	Compliance is a last-minute scramble. Audit trails are incomplete. Change records are manual.	Immutable audit trail from day one. Compliance reports generated automatically. Every change is governed.
AI Integration	AI tools exist but are not integrated into the delivery workflow. Recommendations are siloed.	AI Agent woven through the entire platform: test generation, defect prediction, optimisation, anomaly detection.
Toolchain Fragmentation	Teams spend 30–50% of engineering time on toolchain management, not solution delivery.	Unified Solution Factory: all tools, environments, pipelines, and registries in one governed platform.

2 Problem Space

A rigorous definition of the problems Qala addresses, organised by persona. Each problem is characterised by its context, specific pain, and quantified business impact where available.

2.1 Developer Problems

Developers are the primary builders within any software organisation. Despite being the most critical actors, they spend a disproportionate amount of time fighting infrastructure rather than building solutions.

P-DEV-001 — Environment Inconsistency Across Team

Persona: Developer (all experience levels)

Context: A developer works in a locally configured environment. When code is pushed, it fails CI. When a colleague tries to reproduce a bug, the environment behaves differently.

Pain Point: "Works on my machine" is the default experience. Debugging environment differences takes hours or days per developer per month. New team members spend their first 1–4 weeks setting up their environment.

Business Impact: Industry surveys report 30–50% of engineering time spent on environment and tooling issues rather than product development. Onboarding friction delays productive contribution by weeks.

P-DEV-002 — No Structured Idea-to-Production Pathway

Persona: Developer, founding team, solo builder

Context: A developer has an idea for an application, product, or service. They must manually assemble every tool, workflow, CI/CD pipeline, registry, and deployment mechanism from scratch.

Pain Point: There is no structured pathway from concept to production. Each project starts from a blank slate. Essential capabilities (security scanning, reproducible builds, test management) are skipped under time pressure.

Business Impact: The majority of side projects, internal tools, and early-stage applications never reach production quality. Teams accumulate technical debt from the very first commit due to absent foundation.

P-DEV-003 — Inadequate Visibility into Quality and Health

Persona: Developer, Tech Lead

Context: Code is committed, CI runs, but the developer has no unified view of: code quality trends, test coverage changes, security posture, defect density, or performance trajectory.

Pain Point: Each concern requires a separate tool — a different dashboard for CI, security, code quality, test results, and deployments. There is no single coherent signal of whether the solution is improving or degrading.

Business Impact: Poor visibility leads to quality decay that is only discovered at release time or in production, when fixing issues is most expensive.

2.2 Engineering Organisation Problems

As organisations scale, the coordination, governance, and quality assurance challenges multiply faster than headcount.

P-ORG-001 — Toolchain Proliferation and Fragmentation

Persona: CTO, Engineering Manager, Platform Team

Context: Different teams independently choose CI systems, registries, security tools, test frameworks, and deployment tools. The platform team spends most of its time integrating and maintaining these disparate systems.

Pain Point: No single pane of glass across the engineering organisation. Context switching between tools destroys productivity. Integration work is never-ending. Tool upgrades break integrations unexpectedly.

Business Impact: Gartner estimates that organisations with fragmented toolchains spend 25–40% more per delivered feature than those with unified platforms. Integration maintenance crowds out feature delivery.

P-ORG-002 — Knowledge Loss and Onboarding Cost

Persona: Engineering Manager, Tech Lead, New Developer

Context: When a key engineer leaves, institutional knowledge — architectural decisions, toolchain quirks, incident history, why certain patterns were chosen — is lost. Onboarding new engineers takes months.

Pain Point: Documentation is either absent, outdated, or disconnected from the actual code and configuration. Decision rationale is not captured. Past incidents are not searchable.

Business Impact: The average cost of replacing a mid-level software engineer is estimated at 50–200% of their annual salary, largely driven by lost institutional knowledge and onboarding time.

P-ORG-003 — Cross-Team Quality Inconsistency

Persona: CTO, Engineering Manager

Context: Different teams within the same organisation operate at vastly different quality levels. One team ships with 80% test coverage and zero production incidents; another ships with 20% coverage and weekly rollbacks. There is no visibility or mechanism to address this.

Pain Point: No standardised quality baselines. No cross-team benchmarking. Good practices are not propagated. Poor teams remain poor.

Business Impact: Quality inconsistency creates unpredictable release timelines and customer experiences. High-quality teams are frustrated by being held back by dependencies on low-quality ones.

2.3 Security and Compliance Problems

P-SEC-001 — Security as an Afterthought

Persona: Security Engineer, CISO, Developer

Context: Security review happens at the end of development — in a penetration test before go-live, or worse, after a breach. Developers have no security feedback during development. Vulnerabilities in dependencies are discovered months after introduction.

Pain Point: Security tools exist but are not integrated into the development workflow. Scan results land in a separate system. Remediation is reactive and politically difficult after code is written.

Business Impact: IBM's Cost of a Data Breach Report consistently shows that vulnerabilities discovered in production cost 6–15× more to remediate than those caught during development. Supply chain attacks via compromised dependencies are the fastest-growing attack vector.

P-SEC-002 — Build and Supply Chain Integrity Unverified

Persona: Security Engineer, Compliance Officer, Developer

Context: When an artefact is deployed to production, there is no verifiable record of: what source code produced it, what environment built it, what dependencies it contained, and whether those dependencies were tampered with.

Pain Point: Builds are not hermetic. Dependency integrity is not verified at build time. Build attestations do not exist or are not signed. Supply chain attacks (SolarWinds-class incidents) are undetectable.

Business Impact: High-profile supply chain attacks have cost organisations hundreds of millions of dollars in incident response, regulatory fines, and reputational damage. Without build attestation and hermetic environments, organisations cannot prove compliance with emerging software security standards such as SLSA and SSDF.

P-SEC-003 — Compliance Reporting is Manual and Expensive

Persona: Compliance Officer, CISO, Engineering Manager

Context: Preparing for SOC 2, ISO 27001, or GDPR audits requires weeks of manual evidence collection: pulling audit logs, verifying change records, documenting access control reviews, and assembling proof of security practices.

Pain Point: Evidence is scattered across tools, spreadsheets, and people's email inboxes. Auditors find gaps. Remediation is reactive. Compliance is a one-off project rather than a continuous state.

Business Impact: Mid-market companies spend an average of $500,000–$1,000,000 per year on SOC 2 Type II compliance, largely due to manual evidence collection and remediation. Smaller organisations are effectively locked out of enterprise sales due to compliance barriers.

2.4 Release and Operations Problems

P-REL-001 — Ungoverned, Risky Production Deployments

Persona: Release Manager, Developer, Operations Engineer

Context: Production deployments are manual, undocumented, and inconsistent. There is no enforceable gate checklist. Changes go to production without test sign-off, security clearance, or performance validation. Rollbacks are slow and painful.

Pain Point: Deployments are the highest-risk event in the software lifecycle, yet they are the least governed. Post-deployment incidents are common. Rollback capability is absent or slow. Change records are incomplete.

Business Impact: DORA Research shows that elite performing teams have 208× more frequent deployments with 2,604× faster recovery from failures compared to low-performing teams. The gap is driven almost entirely by governance and automation — not talent.

P-REL-002 — Configuration Drift Causes Unexplained Incidents

Persona: Operations Engineer, Developer, Platform Admin

Context: Production environments silently drift from their defined configurations through manual changes, partial deployments, or undocumented emergency fixes. The next incident is caused by an environment state that no one is aware of.

Pain Point: No mechanism to detect configuration drift. Manual changes are not recorded. Production differs from staging in undocumented ways. Incidents are hard to reproduce and diagnose.

Business Impact: Configuration drift is implicated in an estimated 35–40% of production incidents according to multiple industry surveys. Mean time to diagnosis is extended by hours or days when the root cause is a drifted configuration.

2.5 Industry-Specific and Emerging Use Case Problems

P-IND-001 — No Unified Platform for Non-Software Solution Types

Persona: Physical product manufacturer, CPG company, agricultural business, service business

Context: Professionals building physical goods, packaged products, managed services, or platform businesses have no structured lifecycle management tool. Software tools exist for software. Nothing comparable exists for goods, services, or platforms as first-class managed entities.

Pain Point: Product lifecycle, formulation management, service version control, and platform governance are handled in spreadsheets, email chains, or disconnected ERP systems that were never designed for this purpose.

Business Impact: Recall costs, regulatory fines, and knowledge loss from unmanaged physical product and service lifecycles cost industries billions annually. The root cause is the absence of the same disciplined lifecycle management that software companies take for granted.

P-IND-002 — AI Capabilities Not Integrated into Delivery Workflows

Persona: Developer, ML Engineer, Tech Lead

Context: AI tools for code generation, test writing, defect detection, and anomaly monitoring exist as isolated products. They are not integrated into the delivery workflow. Their outputs are not traceable. Their recommendations are not acted on systematically.

Pain Point: Developers use AI tools in isolation — a code assistant here, a test generator there — but the AI has no context about the solution lifecycle, the history of defects, the quality benchmarks, or the security posture.

Business Impact: The economic value of AI-assisted development is 20–50% productivity improvement according to multiple studies, yet most of this value is captured only in code generation. The much larger opportunity — AI-driven quality, security, and operations — remains largely untapped.

3 Target Market & Persona Analysis

Qala addresses a broad addressable market spanning individual developers to enterprise engineering organisations across every industry that builds solutions.

3.1 Market Segmentation

Segment	Description	Estimated Size	Priority
Developer Tooling (SMB)	Small to medium software companies, SaaS startups, and digital agencies building software products.	~8M organisations globally	P1 — Primary
Enterprise DevOps	Large enterprises operating multiple engineering teams with compliance, audit, and governance requirements.	~200K organisations globally	P1 — Primary
AI/ML Teams	Teams building AI-powered products needing model lifecycle, evaluation pipelines, and cost management.	~500K teams globally	P1 — Primary
Industry Builders	Non-software-native companies building digital solutions in agriculture, manufacturing, CPG, logistics, health, finance.	~2M companies globally	P2 — Secondary
Independent Professionals	Freelancers, hobbyists, enthusiasts, and open-source maintainers building solutions individually.	~25M individuals globally	P2 — Secondary
Public Sector	Government agencies and academic institutions with compliance, auditability, and open data requirements.	~500K agencies globally	P3 — Tertiary

3.2 Persona Deep Dives

Persona	Primary Goals	Key Frustrations	What Success Looks Like
Solo Developer / Indie Builder	Ship a quality product fast. Stay focused on the solution, not the infrastructure.	Setting up CI/CD, registries, and environments from scratch on every project. Security is always skipped under time pressure.	Start a new project and have a full-quality delivery pipeline, hermetic builds, and security scanning running in under 30 minutes.
Startup Engineering Team (2–20)	Iterate rapidly, maintain quality, avoid technical debt, hire and onboard quickly.	Toolchain cobbled together. Every new hire needs 2–4 weeks to onboard. No cross-team visibility into quality.	Any new engineer is productive within a day. Quality metrics are visible to everyone. Releases are calm, not heroic.
Mid-Market CTO / VP Engineering	Predictable delivery, consistent quality across teams, compliance without slowing delivery.	Different teams using different tools. No unified quality view. Compliance is a fire drill. Key engineers	Unified delivery platform. Cross-team quality benchmarking. Compliance artefacts generated

		gatekeeping institutional knowledge.	automatically. Knowledge accessible to all.
Enterprise Platform Engineering Team	Standardise toolchains, enforce governance, enable developer productivity at scale.	Managing hundreds of tool integrations. Drift between environments. Security as a bottleneck. Audit preparation takes months.	Single platform that handles all toolchain governance, security enforcement, and compliance reporting across the enterprise.
Security Engineer / CISO	Shift security left. Ensure build integrity. Enforce policy consistently. Prove compliance.	Security is reactive. No visibility into build provenance. Compliance evidence collected manually. Vulnerabilities discovered late.	Security integrated into every build. Signed attestations for every artefact. Compliance reports generated on demand. Zero-surprise audits.
Industry Solution Builder (non-tech native)	Apply software-quality lifecycle management to physical products, services, and goods.	No tool exists for managing non-software solutions with the same rigour as software.	Formulations, service versions, and platform configurations are governed, auditable, and versioned exactly like software.

4 Opportunity Statements

For every identified problem, Qala's opportunity to deliver measurable value is defined here.

OPP-001 — Eliminate Environment Inconsistency Permanently

Qala's hermetic build system and environment-as-code model eliminate "works on my machine" by ensuring every developer, every CI run, and every deployment uses an identical, cryptographically verified environment. New developers provision a production-equivalent SDE in minutes.

Success Metrics: Onboarding time reduced from weeks to hours. Environment-related CI failures reduced to zero. Developer confidence in build reproducibility: 100%.

OPP-002 — Provide a Structured Idea-to-Production Pathway

Qala provides the complete pathway: Prototype SDE for validation → full developer SDE for building → hermetic CI/CD for delivery → governed release management for production. Every stage is pre-configured, secure by default, and quality-gated.

Success Metrics: Time from idea to first production deployment reduced by 60–80%. Zero security or quality steps skipped due to setup friction. Technical debt from absent foundations eliminated.

OPP-003 — Deliver Unified Quality Visibility Across All Dimensions

A single dashboard surfaces all quality signals in real time: test coverage, defect density, security posture, performance benchmarks, DORA metrics, and AI-generated risk scores. No tool-switching, no manual aggregation.

Success Metrics: Quality degradation caught in development, not production. Mean time to quality diagnosis reduced from days to minutes. Cross-team quality comparison enables targeted improvement.

OPP-004 — Eliminate Toolchain Fragmentation for Engineering Organisations

Qala's Solution Factory model brings all SDEs, pipelines, registries, toolchains, and governance under a single governed platform. Platform teams shift from integration maintenance to value delivery.

Success Metrics: Engineering organisations reduce toolchain-related overhead by 40–60%. Platform engineering team effort shifted from maintenance to strategic improvement. Cross-team standardisation achieved without mandating individual practices.

OPP-005 — Make Enterprise-Grade Security Accessible to Every Builder

Qala integrates SAST, SCA, DAST, container scanning, SEM, secrets management, and SLSA build attestation as first-class platform features — not optional add-ons. Security is embedded at the point where it costs the least to fix.

Success Metrics: Vulnerability discovery shifted 95% to development phase. Supply chain attacks detectable within minutes. Compliance reports generated in seconds. Security accessible to solo developers, not just enterprises.

OPP-006 — Turn Compliance from a Fire Drill into a Continuous State

Qala's immutable audit trail, automated policy enforcement, privacy management pipelines, and compliance report generation make SOC 2, ISO 27001, GDPR, CCPA, and SLSA compliance a continuous operational state rather than a periodic project.

Success Metrics: Annual compliance preparation time reduced from weeks to hours. Audit findings reduced by 80%+. Compliance accessible to mid-market companies that previously could not afford it.

OPP-007 — Apply Software Lifecycle Rigour to All Solution Types

Qala's solution model treats Applications, Systems, Goods, Products, Services, and Platforms as first-class entities with governed version control, release management, and change control. Physical product manufacturers, CPG companies, and service businesses can apply the same discipline.

Success Metrics: Non-software-native businesses reduce time-to-recall identification from days to minutes. Regulatory compliance for formulation changes governed automatically. Platform governance achievable without engineering teams.

OPP-008 — Integrate AI Intelligence Throughout the Delivery Lifecycle

Qala's AI Agent is not a standalone tool but an intelligence layer woven through every platform domain: test generation, defect prediction, optimisation recommendations, anomaly detection, and cross-team learning. All AI actions are traceable, auditable, and improvable.

Success Metrics: Developer productivity gains of 20–40% from AI-assisted quality and security. Defect escape rate reduced by AI risk scoring. Platform continuously improves from aggregated learning across all users.

5 Problem-Solution Fit Matrix

Mapping every identified problem to the specific Qala capability that resolves it, with the primary success metric.

Problem ID	Problem	Qala Capability	Primary Success Metric
P-DEV-001	Environment inconsistency	Hermetic SDEs + environment-as-code	Zero environment-related CI failures
P-DEV-002	No idea-to-production pathway	Prototype SDE → full SDE → CI/CD → Release	80% reduction in time from concept to production
P-DEV-003	No unified quality visibility	Unified quality dashboard + AI risk scoring	All quality signals in one portal, updated in real time
P-ORG-001	Toolchain fragmentation	Solution Factory with shared toolchains	40–60% reduction in toolchain overhead
P-ORG-002	Knowledge loss + onboarding cost	Versioned artefacts, AI semantic search, history	New developer productive within 1 day
P-ORG-003	Cross-team quality inconsistency	Cross-project benchmarking + AI cross-learning	Cross-team quality gap reduced by 50% in 12 months
P-SEC-001	Security as afterthought	SAST/SCA/DAST embedded in every CI pipeline	95%+ of vulnerabilities caught in development phase
P-SEC-002	Build integrity unverified	Hermetic builds + SLSA attestation + signing	100% of production artefacts carry verified attestations
P-SEC-003	Compliance reporting is manual	Automated audit trail + compliance report gen	Compliance preparation time reduced to under 4 hours
P-REL-001	Ungoverned production deployments	8-gate release management + governed deployment	Change failure rate reduced by 70%; MTTR under 5 minutes
P-REL-002	Configuration drift	Drift detection + CM with CCR workflow	All configuration drift detected and alerted within 15 minutes
P-IND-001	No platform for non-software types	Solution model: Goods, Products, Services	Non-software lifecycle managed at parity with software
P-IND-002	AI not integrated into delivery	AI Agent woven through all platform domains	30%+ reduction in defect escape rate; AI recommendation accept rate > 60%

6 Success Metrics & Outcomes

How Qala measures that problems are genuinely solved — not just that features are shipped.

6.1 Developer Experience Metrics

Metric	Baseline (Industry Average)	Qala Target
Time to provision new development environment	2–14 days (manual setup)	< 10 minutes
Onboarding time to first productive commit	1–4 weeks	< 1 business day
Environment-related CI failures per developer/month	5–15 failures	0 failures (hermetic)
Time spent on toolchain vs. product work	30–50% of engineering time	< 10% of engineering time
Defect discovery phase: development vs. production	~40% in dev, ~60% in prod	> 95% in development phase

6.2 Quality & Delivery Metrics (DORA)

DORA Metric	Industry Average	Qala Elite Target
Deployment frequency	Once per month to once per week	Multiple deploys per day
Lead time (commit to production)	1 week to 1 month	< 1 day
Change failure rate	15–45%	< 5%
MTTR (production incident)	1 day to 1 week	< 1 hour (5-minute rollback)

6.3 Security Metrics

Metric	Baseline	Qala Target
Vulnerability discovery phase	50%+ post-deployment	< 5% post-deployment
Mean time to detect threat in SDE	Days to never	< 90 seconds (automated SEM)
Build attestation coverage	0% (industry average)	100% of production artefacts
Compliance audit preparation time	4–12 weeks	< 4 hours
Dependency integrity verified at build	< 10% of organisations	100% via digest pinning

6.4 Platform Adoption Metrics

Metric	Target at 12 Months	Target at 36 Months
Active SDEs on platform	10,000	500,000
Monthly active developers	5,000	250,000
Industries represented	10+	25+
AI recommendation acceptance rate	> 60%	> 75%
Automated compliance report generation	> 80% of audit evidence automated	> 99%
Average platform NPS	> 50	> 65

7 Constraints, Assumptions & Risks

7.1 Constraints

Constraint	Description
Platform Independence	Qala must support multi-cloud deployment and not depend on proprietary infrastructure from any single cloud provider.
Backwards Compatibility	API backwards compatibility must be maintained within a major version. Breaking changes require a major version with a minimum 6-month deprecation notice.
Open Standards	Qala must implement open standards: OpenTelemetry for observability, SLSA for build attestation, SPDX/CycloneDX for SBOM, OAuth2/OIDC for identity, and OpenAPI for all service interfaces.
Data Sovereignty	All tenant data must be processable and storable within a designated geographic region; cross-region data transfers must be explicitly opt-in and auditable.
Offline/Air-Gap Support	Edge SDE deployments must support operation in environments with no internet connectivity. Dependency caches must be pre-populatable.
Performance Baselines	Platform control plane must maintain 99.9% monthly uptime. All API responses must be < 500ms at p95 under normal load.

7.2 Assumptions

Assumption	Basis
Hermetic builds are technically feasible for all target languages	Validated for Rust, Go, Scala, Java, Python, Node.js, and .NET via container-based build isolation.
Cross-industry solution model generalises sufficiently	Solution model (Application/System/Good/Product/Service/Platform) has been validated against 25+ industry use cases.
AI model quality will improve over time	AI Agent capabilities will compound as training data accumulates from platform usage. Initial release may have lower accuracy on less common patterns.
Developers will accept hermetic constraints	Hermetic environment requirements may initially feel restrictive to developers accustomed to ad-hoc environments. Tooling UX must compensate.
Compliance frameworks will not change materially	SOC 2, ISO 27001, GDPR, CCPA, and SLSA requirements are stable. Regulatory changes will require compliance module updates.

7.3 Risks

Risk	Likelihood	Impact	Mitigation
Toolchain complexity overwhelms new users	High	High	Invest in opinionated defaults, guided onboarding, and curated templates for common use cases.

AI recommendations of poor quality erode trust	Medium	High	Track acceptance rates. Require human review for all AI-generated actions. Invest in feedback loops.
Platform lock-in perception deters adoption	Medium	High	Adopt open standards for all data formats. Provide import/export for all entities. Never use proprietary formats.
Performance at scale exceeds architecture limits	Low	Critical	Architect for horizontal scaling from day one. Load-test at 10× expected peak before launch.
Regulatory changes break compliance modules	Medium	Medium	Modular compliance framework. Rapid update cycle for compliance reports. Regulatory monitoring capability.
Supply chain attack on Qala platform itself	Low	Critical	Eat our own cooking: Qala platform is built with Qala hermetic builds and SLSA attestations.
Fragmented adoption (some teams, not all)	High	Medium	Factory-level enforcement makes partial adoption self-defeating. Leadership sponsorship strategy.

8 Prioritised Problem Backlog

Problems are prioritised by the combination of: severity of impact to the target persona, breadth of market affected, and Qala's distinctive advantage in solving it.

Rank	Problem ID	Problem	Severity	Breadth	Qala Advantage	Priority
1	P-SEC-002	Build & supply chain integrity unverified	Critical	Universal	Hermetic builds + SLSA are core architecture, not bolt-on	P1
2	P-DEV-001	Environment inconsistency	High	Universal	SDE model eliminates this class of problem entirely	P1
3	P-REL-001	Ungoverned production deployments	High	Universal	8-gate release management built into platform primitives	P1
4	P-SEC-001	Security as afterthought	High	Universal	Security integrated at every layer, not optional	P1
5	P-ORG-001	Toolchain fragmentation	High	Enterprise	Solution Factory model uniquely solves this at the factory level	P1
6	P-DEV-002	No idea-to-production pathway	High	Universal	Prototype SDE to production is a first-class Qala workflow	P1
7	P-REL-002	Configuration drift	Medium	Universal	Environment-as-code + continuous drift detection built-in	P1
8	P-SEC-003	Compliance reporting is manual	High	Enterprise	Immutable audit trail + automated compliance reports from day one	P2
9	P-DEV-003	No unified quality visibility	Medium	Universal	Single dashboard across all quality domains via unified event bus	P2
10	P-ORG-002	Knowledge loss + onboarding cost	Medium	Universal	Versioned artefacts + AI retrieval make knowledge ambient	P2
11	P-IND-002	AI not integrated into delivery	Medium	Growing	AI Agent is a first-class platform subsystem, not a sidebar	P2
12	P-ORG-003	Cross-team quality inconsistency	Medium	Enterprise	Factory-level benchmarking and AI cross-team learning	P2
13	P-IND-001	No platform for non-software solution types	Medium	Niche	Solution model generalises beyond software by design	P3

End of Document — Qala Solution Factory OS | Problem Requirements Document v1.0